Can Agentic Trading Systems Pay for Their Own Intelligence?

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Abstract

Large language model (LLM) agents are increasingly used in trading systems, where model reasoning, tool use, and continual decisions incur costs that are expected to produce trading value. Existing evaluations typically report performance metrics, but rarely examine ***agentic viability***: whether dynamic LLM-mediated decisions convert their induced costs into measurable incremental profit. To apply this criterion, we introduce **TradeLens**, a trace-grounded diagnostic toolkit for evaluating agentic trading systems from their trading records, runtime traces, and deployment configurations. It reconstructs trading trajectories, attributes profit and cost to interpretable evidence, and diagnoses whether and why an agent pays for its own intelligence. We conduct extensive analysis across backbone models, capital scales, trading frequencies, and system architectures, together with deployment discussion. Our results show that viability hinges on intelligence-to-profit conversion: models exhibit different failure patterns, such as poor asset selection in DeepSeek-V3.2 and negative timing in GLM-4.7, while capital scale, trading frequency, and architecture matter primarily in these runs, especially through their effects on decision-attributed timing value. These findings reframe the evaluation of LLM-based trading agents from capability-centric performance ranking to trace-grounded diagnosis of intelligence-to-profit conversion. Our code is available at https://github.com/ParadooxAI/TradeLens.

1 Introduction

With strong general reasoning abilities (Yu et al., 2024a; Liu et al., 2023a; Guo et al., 2025a) and increasing adaptation to financial tasks (Wu et al., 2023; Chen et al., 2026a; Liu et al., 2023b; Zhang et al., 2026b; Liu et al., 2025), LLM agents are moving from financial question answering to direct participation in trading workflows.

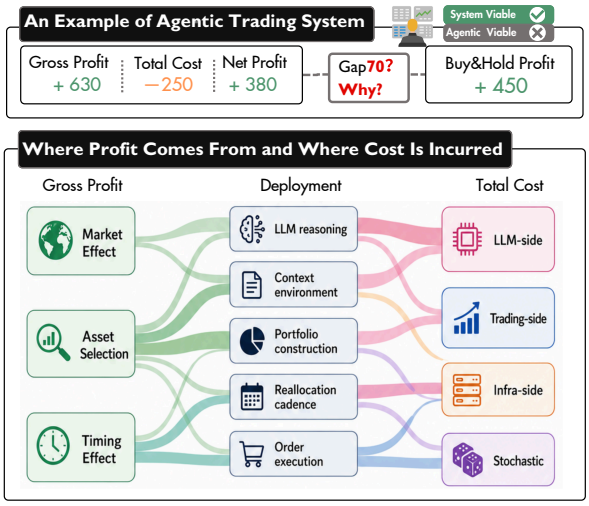


Figure 1: Motivation. Profitability does not necessarily imply positive agentic value. Our framework diagnoses why by disentangling the deployment drivers that jointly shape returns and costs.

Recent systems use LLMs as trading copilots (Zhou et al., 2024; Guo and Shum, 2025), portfolio managers (Zhao et al., 2025; Li et al., 2026), and market-analysis agents equipped with external tools (Han et al., 2025; Zhang et al., 2026a,c).

Observation. Despite promising results, a critical gap remains in how agentic trading systems are evaluated. Most existing studies report *gross profit* or metrics such as Sharpe ratio (Li et al., 2025b; Chen et al., 2026b; Xie et al., 2023). While useful for measuring trading performance, these metrics are incomplete for deployment: developers must assess not only how much profit a system makes, but also where the profit comes from and how deployment costs affect the realized margin. This echoes recent cost-aware LLM evaluations balancing monetary costs and reasoning efficiency (Erol et al., 2026; Zellinger and Thomson, 2025; Wang et al., 2025b; Zhang et al., 2025), and whether AI expenditure ultimately yields measurable economic value (Howell et al., 2023; NANDA, 2025).

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However, this question becomes more stringent in agentic way: cost is the price paid for LLM-mediated intelligence, including model reasoning, tool use, memory retrieval, and continual decision-making. Beyond operating within a budget (Wang et al., 2024), the system should also meet the *token-economy requirement* (Chen et al., 2026c): generating enough incremental profit to justify the intelligence consumed by its trading decisions.

The blind spot. This requirement creates a profit-side and cost-side double blind in evaluating agentic trading systems. As shown in Figure 1, an agentic system may earn a gross profit of 630, but after 250 in deployment costs, its net profit drops to 380; meanwhile, a simple buy-and-hold baseline without LLM inference or active trading costs may earn 450 over the same period. Thus, gross profit can overstate decision value when the agent does not outperform passive market exposure, and can overstate deployability when end-to-end costs absorb the realized margin. We therefore ask: *Can agentic trading systems pay for their own intelligence, and how can we diagnose whether LLM-mediated decisions convert their induced costs into incremental trading value?*

Call for rigorous assessment. Answering this question requires identifying both the value added by LLM intelligence and the costs it incurs. The decomposition in the lower panel of Figure 1 shows that both profit and cost are shaped by the deployment process. A profitable trajectory may mainly reflect market movement or favorable initial asset selection, while the agent's dynamic decisions contribute little or even reduce value. Conversely, higher reasoning, tool-use, or execution costs are not necessarily wasteful if they produce sufficient active profit. This makes rigorous assessment necessary for diagnosing whether LLM-mediated decisions are economically justified.

Challenge of trace-grounded diagnosis. However, performing this diagnosis is difficult because the evidence needed to evaluate LLM-mediated decisions is fragmented across different records. Portfolio trajectories show what the system earned, but not whether the gain came from passive exposure, initial allocation, or dynamic agent intervention. Runtime traces show what the agent did, including model calls, context processing, tool use, retries, and trading actions, but not whether these activities improved asset selection, timing, or portfolio value. Deployment costs further vary with model choice, context length, tool-use frequency, trading cadence, and execution behavior, making a single cost proxy insufficient. Therefore, the challenge is to ground the diagnosis of intelligence-to-profit conversion in heterogeneous evidence from trading outcomes, runtime traces, and deployment costs.

Solution. To address these challenges, we propose **TradeLens**, a trace-grounded diagnosis toolkit for evaluating whether agentic trading systems can pay for their own intelligence. This toolkit first reconstructs the investment process from portfolio records, runtime traces, and deployment configurations, and attributes profit and cost to interpretable components. Based on this evidence, we define two viability notions (Section 3): *system viability*, which asks whether the deployed system pays for itself after end-to-end costs, and *agentic viability*, which asks whether dynamic LLM-mediated decisions create enough active value to justify their decision-induced costs. Then, an LLM-based diagnosis agent generates evidence-grounded explanations and strategy-level suggestions from the observed trading trajectory, and converts these diagnoses into visual and textual reports (Section 4). We evaluate **TradeLens** through controlled experiments and practitioner feedback, showing that it reveals failure modes hidden by system's profitability and helps users assess deployability and identify revision directions (Section 5).

Collectively, we reframe the evaluation of agentic trading systems from capability-centric to economically grounded. To summarize, we make three contributions:

(i) We formulate *profit–cost viability* as a trace-grounded evaluation problem for agentic trading systems, distinguishing whether the overall deployed system pays for itself and whether LLM-mediated decisions justify their induced costs.

(ii) We build **TradeLens**, an open-source audit toolkit that operationalizes the methodology using trading records, runtime traces, and deployment configurations, and produces evidence-grounded diagnostic reports.

(iii) We conduct empirical viability studies across backbone model, capital scale, trading frequency, and architecture, together with practitioner feedback, showing how profit sources and intelligence cost jointly determine deployability.

2 Related Work

2.1 LLM-Based Trading Agents

LLM-based trading agents have expanded from financial analysis and signal generation (Wang et al., 2025a; Xing, 2024) to decision pipelines that use memory, tools, and multi-agent coordination (Yu et al., 2024b, 2025; Xiao et al., 2025; Li et al., 2025c). Recent work also moves evaluation closer to practical trading settings, including cryptocurrency markets (Li et al., 2024; Luo et al., 2026), portfolio construction (Guo et al., 2025b; Zhao et al., 2025), streaming interaction (Li et al., 2025a; Fan et al., 2025), and end-to-end research platforms (Sun et al., 2023; Zhang et al., 2026d). Across these works, evaluation mainly focuses on capability or gross trading performance, leaving profit sources and economic viability underexplored. **TradeLens** addresses this gap as a diagnosis toolkit for existing trading agents, checking whether these agents can pay for their intelligence.

2.2 Cost-Aware and Trace-Grounded AI Evaluation

Recent work increasingly incorporates computational cost into model evaluation. Cost-of-pass estimates the monetary cost of obtaining a successful answer (Erol et al., 2026; Zellinger and Thomson, 2025); efficient-reasoning studies examine thinking style and redundancy in agentic systems (Wang et al., 2026; Lin et al., 2026; Wang et al., 2025b; Zhang et al., 2025); and other work advances resource-constrained evaluation (Chen et al., 2024; Huang et al., 2025; Liu et al., 2026; Yang et al., 2026; Zhang et al., 2026e,f). These studies mainly treat cost as a resource-efficiency variable. For LLM agents, however, costs are also induced by the decision process itself, including reasoning, tool use, memory access, and multi-step interaction (Shinn et al., 2023). Token economics further frames these activities as paid token and compute consumption before downstream actions are produced (Wang et al., 2024; Chen et al., 2026c). Prior work on agent evaluation and diagnosis has shown that such trajectories provide useful evidence beyond final task outcomes (He et al., 2026; Ou et al., 2025). In trading workflows, we use this trace-grounded view in a different setting: agent activities can be aligned with resource use, trading actions, and portfolio outcomes, enabling diagnosis of whether decision-induced costs are converted into trading value.

2.3 Performance Attribution in Financial Evaluation

Financial evaluation has long separated raw return from interpretable sources of return. Risk-adjusted metrics such as the Sharpe ratio (Sharpe, 1994) and regime-aware analyses (Ang and Timmermann, 2012) show that portfolio outcomes can reflect volatility or market conditions rather than decision skill. Classical performance attribution decomposes benchmark-relative returns into allocation, selection, and interaction effects (Brinson et al., 1986, 1991), with later work using attribution to study active management skill and dynamic allocation behavior (Grinold and Kahn, 2000; Hsu et al., 2010; Al-Aradi and Jaimungal, 2018).

Recent financial AI benchmarks and trading-agent studies often report predictive accuracy, cumulative return, or portfolio-level profit (Ko and Lee, 2024; Li et al., 2025b; Chen et al., 2026b; Fan et al., 2025). These metrics are useful, but they do not explain whether profit comes from market exposure, asset selection, or dynamic reallocation. **TradeLens** adopts a lightweight attribution view for this purpose and joins it with system-cost accounting, so viability is assessed from decision-attributed margin rather than gross profit alone.

3 Methodology: Profit–Cost Viability

This section defines how we evaluate whether an agentic trading system can pay for the intelligence it consumes. We focus on two questions: whether the system is economically viable, and whether its agentic decisions generate sufficient incremental value to justify their induced costs.

3.1 Profit and Cost Attribution

Given an evaluation window $[1, T]$, let V_0 be the initial portfolio value and V_T^{dyn} be the realized terminal value of the deployed agentic system. The cumulative gross profit is

$$P_{1:T} = V_T^{\mathrm{dyn}} - V_0. \tag{1}$$

Profit Attribution. Classical performance attribution decomposes portfolio returns into interpretable sources before evaluating investment decisions (Brinson et al., 1986, 1991). We adopt this logic for agentic trading systems, but reformulate it at the portfolio-value level. For diagnosis, we separate gross profit using two counterfactual baselines over the same asset universe and evaluation.

Let V_T^{sys} denote the terminal value of a passive market benchmark based on the system's initial market exposure, and let V_T^{base} denote the terminal value obtained by holding the system's first active allocation unchanged. The comparison between these baselines separates market exposure from asset-selection value, while the remaining difference between the realized dynamic path and the static-allocation path is attributed to timing:

$$\begin{aligned} P^{\text{sys}} &= V_T^{\text{sys}} - V_0, \\ P^{\text{asset}} &= V_T^{\text{base}} - V_T^{\text{sys}}, \\ P^{\text{timing}} &= V_T^{\text{dyn}} - V_T^{\text{base}}. \end{aligned} \tag{2}$$

Thus,

$$P_{1:T} = P^{\text{sys}} + P^{\text{asset}} + P^{\text{timing}}. \tag{3}$$

Here, P^{sys}, P^{asset}, and P^{timing} correspond to market exposure, asset selection, and timing decisions, respectively. In our analysis, P^{timing} is treated as the profit component most directly attributable to agentic intervention.

Cost Attribution. Transaction cost economics treats market participation as non-frictionless: information processing, decision-making, coordination, and execution all consume resources (Williamson, 2005). For agentic trading systems, "intelligence" is produced through paid inference and then realized through execution and deployment. We decompose the deployment cost in each period t as

$$C_t = C_t^{\text{llm}} + C_t^{\text{trd}} + C_t^{\text{inf}} + C_t^{\text{sto}}, \tag{4}$$

where C_t^{llm} is LLM inference cost, C_t^{trd} is trading execution cost, C_t^{inf} is infrastructure and data-access cost, and C_t^{sto} captures residual hard-to-enumerate costs. The cumulative cost is

$$C_{1:T} = \sum_{t=1}^{T} C_t. \tag{5}$$

Detailed profit and cost itemization is provided in Appendix B and Appendix C.

3.2 Viability Criteria

Viability is evaluated over a common decision window $[1, T]$. We then distinguish two levels of viability. System viability asks whether the whole pipeline pays for itself, whereas agentic viability asks a more specific agent-evaluation question: whether dynamic LLM intervention creates enough incremental value to justify the costs it induces.

System viability. System viability asks whether the fully deployed pipeline pays for itself. The system-level net profit is

$$R_{1:T} = P_{1:T} - C_{1:T}. \tag{6}$$

The system is viable if

$$\text{SystemViable} = \mathbb{I}[R_{1:T} \geq 0]\,. \tag{7}$$

Agentic viability. Agentic viability asks whether dynamic agent intervention itself is economically justified. It removes passive market exposure (P^{sys}) and initial asset-selection effects (P^{asset}), and compares timing profit against costs that are directly induced by dynamic decisions. We classify costs that scale with token usage, retries, and tool calls as dynamic costs, and treat fixed hosting as static costs. The dynamic cost is

$$C_{1:T}^{\text{dyn}} = \sum_{t=1}^{T} \left(C_t^{\text{llm}} + C_t^{\text{trd}} + C_t^{\text{sto}} \right), \tag{8}$$

excluding infrastructure cost which are treated as system-level deployment costs. The net agentic value is

$$R_{1:T}^{\text{agent}} = P^{\text{timing}} - C_{1:T}^{\text{dyn}}. \tag{9}$$

The agent is viable if

$$\text{AgenticViable} = \mathbb{I}\left[R_{1:T}^{\text{agent}} \geq 0 \right]. \tag{10}$$

4 TradeLens

In this section, we describe how **TradeLens** implements this methodology by reconstructing trading trajectories, aggregating runtime costs, and generating structured diagnostic reports.

4.1 Overview

The design goal of **TradeLens** is to make agentic trading systems auditable without assuming a specific agent architecture. Different systems may use different LLM backbones, prompting strategies, tool chains, memory modules, or multi-agent designs. However, once deployed, they tend to leave similar observable traces. **TradeLens** therefore defines a trace interface around the following three inputs: *trading results*, *runtime traces*, and *system configurations*.

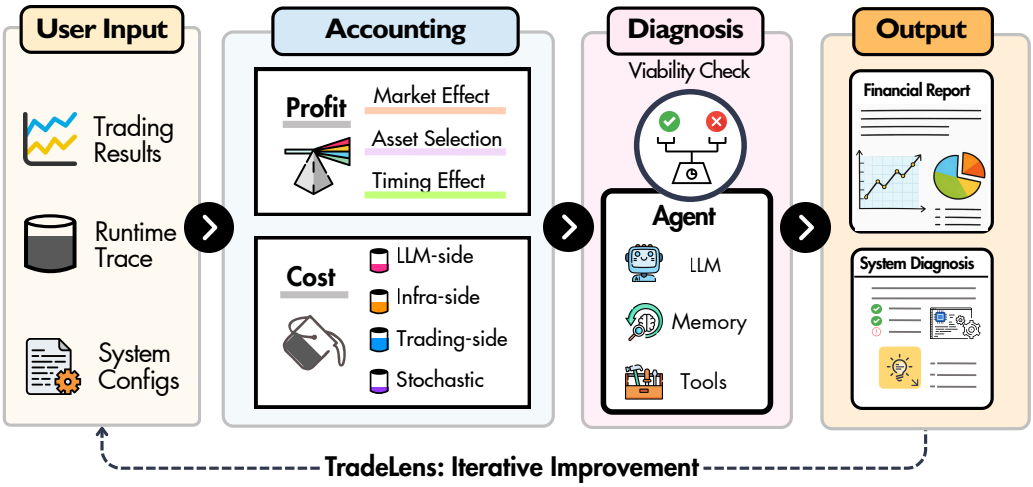


Figure 2: **The pipeline overview.** Retail traders can use the toolkit to evaluate the profit–cost viability of their agentic systems under realistic assumptions using trading results, runtime traces, and basic configurations. The diagnostics support an iterative loop of system adjustment and re-evaluation.

Input. Trading results describe what happened in the market-facing process, including portfolio values, position weights, executed orders, and transaction logs, used to reconstruct the realized portfolio trajectory. Runtime traces describe how the system produced its decisions, including model usage, tool calls, retries, latency, and execution events, used to recover cost drivers and system activities. System configurations describe the deployment setting, including the LLM backbone, capital scale, trading frequency, market window, cost assumptions, and agent architecture. They provide the context needed to interpret both profit and cost.

Outputs. TradeLens produces two user-friendly reports (see format in Appendix A). The financial report summarizes portfolio performance, profit attribution, cost attribution, and viability status. The agentic system diagnosis report explains the failure mode and provides evidence-grounded revision suggestions for the trader.

4.2 Accounting Layer

As shown in Figure 2, the accounting layer traces the trading trajectory and applies the attribution method defined in Section 3 to profit and cost. The profit module replays trading actions to recover portfolio values and decomposes gross profit into market, selection, and timing effects. The cost module accounts for commissions, LLM usage, infrastructure, data subscriptions, and uncertainty costs. Eventually, it delivers auditable results, including the action ledger, attribution tables, and diagnostic figures, providing fixed evidence for the next layer.

4.3 Diagnosis Layer

The diagnosis layer converts accounting results into interpretable system-level diagnoses. It checks viability, identifies dominant failure modes, and generates diagnostic reports. It first extracts key metrics from the audit results and summarizes runtime records into a compact execution profile. These structured inputs are then combined under a constrained output contract, requiring the final report to follow a fixed diagnostic structure and to ground conclusions in supplied evidence.

The diagnosis produces two types of analysis. First, it interprets viability by combining trading outcomes and runtime measurements, identifying whether weak performance stems from insufficient trading profit, high model-side cost, excessive decisions, or execution frictions. Second, it maps trace-supported failure modes to actionable revisions. For example, frequent decisions with limited incremental profit indicate a cadence–return mismatch. The agent then prioritizes operational fixes, such as reducing redundant model calls or decision frequency, and structural revisions, such as improving execution logic, portfolio construction, trading architecture, or the LLM's role in the decision loop.

5 Experiments

We empirically evaluate the economic viability of agentic trading systems through cost-profit breakdowns and conduct experiments on four deployment dimensions: backbone model, capital scale, trading frequency, and system architecture. We address the following research questions (RQs):

- **RQ1 (Backbone model)**: How do LLM backbones differ in profit sources, intelligence cost, and net margin under the same trading pipeline?
- **RQ2 (Capital scale)**: Does capital scaling improve agentic viability by diluting fixed costs, or does it amplify the value and risk of LLM-mediated decisions?
- **RQ3 (Trading frequency)**: Does higher decision frequency create enough marginal profit to cover additional inference and execution costs?
- **RQ4 (System architecture)**: Does architectural complexity translate into decision-attributed gains, or merely increase coordination and execution costs?

5.1 Setup

Deployment setting. We instantiate **TradeLens** on top of a representative agentic trading system, AI-Trader (Fan et al., 2025), to evaluate whether the toolkit can audit realistic decision pipelines. The system trades a fixed universe of liquid U.S. equities over a two-month market window from December 1, 2025 to January 30, 2026, starting with an initial capital of $100,000. Market data are accessed through licensed provider APIs/subscriptions, and raw provider data are not redistributed.

Profit–cost setting. We adopt a simplified but end-to-end cost formulation that covers four deployment-relevant components: LLM-side decision cost, trading-side execution cost, infrastructure cost, and stochastic runtime overhead. On the profit side, we construct two attribution baselines: a systematic exposure baseline based on the S&P 500 return over the same window, applied only to the portfolio's initially invested capital, and a static allocation baseline that holds the initial portfolio weights unchanged. Detailed settings are reported in Appendix D.1.

Experiment scenarios. For backbone model (RQ1), we compare flagship LLMs from leading providers, including GPT-5.2 (OpenAI, 2025), Gemini 3 Flash (Google, 2025), Claude Sonnet 4.5 (Anthropic, 2025), Qwen3-Max (Yang et al., 2025), DeepSeek-V3.2 (DeepSeek-AI, 2025), GLM-4.7 (Z.AI, 2025), Kimi-K2 (Team et al., 2026), MiniMax-M2.1 (MiniMax, 2025), Llama-4-scout (Meta AI, 2025), and Mistral-Large-3 (Mistral AI, 2025). For RQ2–RQ4, we utilize GPT-5.2 and DeepSeek-V3.2 as representative backbones.

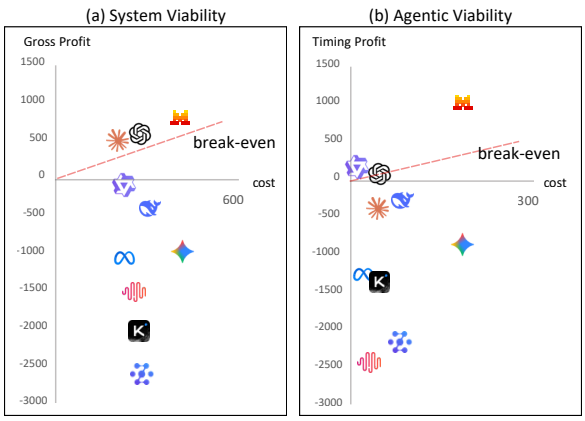


Figure 3: **Viability across backbone models.** (a) System viability and (b) agentic viability: profit (y-axis) versus cost (x-axis) for 10 backbone LLMs under identical trading configurations.

RQ2 varies the initial endowment from $10,000 to $500,000 and reports net profit, cost-to-capital ratio, and break-even status. RQ3 compares daily and hourly decision frequencies using marginal profit minus marginal cost in one month horizon. RQ4 explores the architecture reasoning complexity by placing AI-Trader as a moderate design. We utilize the typical Chain-of-Thought prompting (See details in Appendix D.4) and DeepFund (Li et al., 2025a) as simple and complex designs, respectively. All variants use the same trading logic and execution assumptions.

5.2 Results Analysis (RQ1–RQ4)

Our analysis focuses on viability patterns across the evaluated configurations and market windows.

RQ1 (Backbone model). The key difference across LLM backbones lies less in their operating cost than in how they generate, or fail to generate, active profit. Figure 3 and Table 1 reveal models viability levels to four groups.

First, GPT-5.2 and Mistral-Large-3 are viable at both system and agentic levels. Mistral-Large-3 achieves the strongest agentic profit despite relatively high cost, with a large positive timing effect suggesting dynamic reallocation as a candidate driver relative to the static baseline.

Second, Claude Sonnet 4.5 represents a system-viable but agentically weak backbone. It achieves positive net profit but negative agentic profit, indicating that the system covers costs while the dynamic component adds no incremental value. This case motivates evaluating system and agentic viability separately.

Metric	DeepSeek-V3.2	MiniMax-M2.1	Kimi-K2	GPT-5.2	GLM-4.7	Llama-4-scout	Qwen3-Max	Claude Sonnet 4.5	Gemini 3 Flash	Mistral-Large-3
Gross Profit [(a)]	-388.29	-1474.62	-1975.63	505.76	-2554.81	-1020.38	-50.43	519.40	-975.58	777.81
Market Effect	1351.45	1718.95	1714.40	1535.80	1696.78	1521.73	0.00	1716.01	1351.45	1719.04
Asset Selection	-1500.44	-813.13	-2399.00	-1125.69	-2099.53	-1304.89	-175.90	-870.58	-1500.44	-1981.37
Timing Effect [(1)]	-239.30	-2380.44	-1291.03	95.65	-2152.06	-1237.22	125.46	-326.02	-826.59	1040.14
Total Cost [(b)]	304.94	252.80	275.67	258.79	296.95	228.90	223.04	255.04	422.38	414.82
Trading [(2)]	63.00	33.00	56.00	32.00	79.38	10.00	4.00	23.00	200.00	195.00
LLM [(3)]	23.84	1.35	2.06	9.03	0.83	0.04	0.90	11.38	4.34	1.50
Infra	208.20	208.20	208.20	208.20	208.20	208.20	208.2	208.20	208.20	208.20
Stochastic [(4)]	9.90	10.25	9.42	9.56	8.55	10.66	9.93	12.47	9.84	10.11
Net Profit [(a-b)]	-693.23	-1727.42	-2251.30	246.97	-2851.76	-1249.29	-273.47	264.36	-1397.96	362.99
Agentic Profit [(1-2-3-4)]	-336.04	-2425.04	-1358.50	45.06	-2240.81	-1257.92	110.63	-372.86	-1040.76	833.52

Table 1: Viability across backbone models. Net profit measures system viability, while agentic profit measures whether timing value covers decision-induced costs. Market Effect applies only to initially invested capital; a zero value indicates that all capital was initially held in cash.

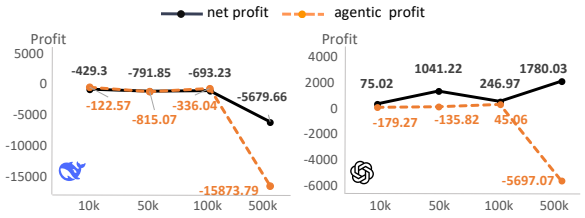


Figure 4: **Viability across capital scale.** *Net Profit* and *Agentic Profit* under varying initial cash investment scale for GPT-5.2 and DeepSeek-V3.2.

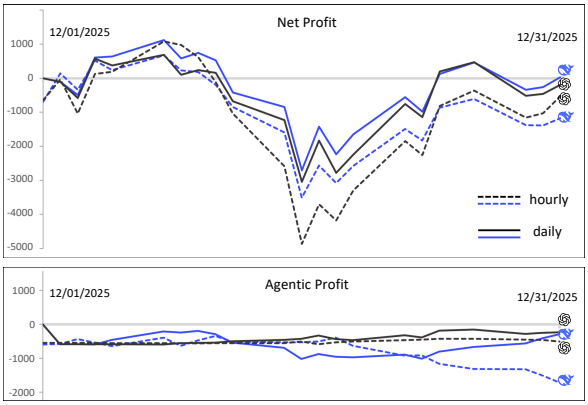


Figure 5: **Viability across trading frequency.** Cumulative net profit and agentic profit over time under hourly (dashed) and daily (solid) trading frequencies for GPT-5.2 and DeepSeek-V3.2. The horizontal line at zero indicates the break-even line.

Third, Qwen3-Max is agentically viable but system-non-viable: its positive agentic profit is insufficient to offset total system cost.

Fourth, the remaining six backbones are non-viable at both levels in the observed window. DeepSeek-V3.2 is dominated by negative selection, while GLM-4.7 and MiniMax-M2.1 show larger negative timing effects.

Overall, backbone choice changes not only the magnitude of profit and cost, but also the source of economic value and failure.

RQ2 (Capital scale). Figure 4 shows that capital scaling does not uniformly improve viability. DeepSeek-V3.2 exhibits a negative pattern. Both system and agentic viability remain negative across all scales, with losses expanding sharply at 500k. These losses are mainly associated with unfavorable timing effects rather than operating costs. For GPT-5.2, system viability remains positive but varies non-monotonically with capital scale, whereas agentic viability is also non-linear. However, with 500k initial cash, it only gains a modest return compared to the setting with 50k, and this result is mainly due to passive market exposure, while the timing effect remains negative. The diagnosis therefore recommends streamlining the inference and order pipeline to reduce latency and slippage, and recover timing value.

Overall, capital scale acts as more than a simple cost-dilution mechanism, as it may amplify both favorable timing outcomes and strategy-level losses. See Appendix D.2 for more details.

RQ3 (Trading frequency). Figure 5 compares cumulative net profit and agentic profit under hourly and daily decision frequencies. The results show that higher decision frequency does not improve viability. For both backbones, daily trading outperforms hourly trading in terms of both net profit and agentic profit. DeepSeek-V3.2 improves from a net profit of -1222.24 under hourly trading to 29.84 under daily trading, while GPT-5.2 improves from -585.46 to -198.28. This stems not merely from higher operating costs, but from degraded active-decision quality under the hourly setting.

Hourly mode raises total cost for both and might produce worse gross profit, especially for DeepSeek-V3.2 (see Appendix Table 6). This suggests that more frequent decisions introduce additional trading noise and timing errors.

Metric	DeepSeek-V3.2			GPT-5.2		
	CoT	AI-Trader	DeepFund	CoT	AI-Trader	DeepFund
Gross Profit [(a)]	-1236.25	-388.29	610.00	341.87	505.76	881.25
Market Effect	514.70	1351.45	150.25	1432.69	1535.80	1338.89
Asset Selection	1463.8	-1500.44	-56.88	-518.61	-1125.69	-902.11
Timing Effect [(1)]	-3214.75	-239.30	516.62	-572.20	95.65	444.47
Total Cost [(b)]	278.57	304.94	397.02	245.44	258.79	403.11
Trading [(2)]	60.00	63.00	174.00	25.23	32.00	178.00
LLM [(3)]	1.35	23.84	3.94	1.27	9.03	6.13
Infra	208.20	208.20	208.20	208.20	208.20	208.20
Stochastic [(4)]	9.02	9.90	10.67	10.75	9.56	10.59
Net Profit [(a-b)]	-1514.82	-693.23	212.98	96.43	246.97	478.14
Agentic Profit [(1-2-3-4)]	-3285.12	-336.04	328.01	-609.45	45.06	249.76

Table 2: Viability across system architecture with $100,000 initial cash.

RQ4 (System architecture). Table 2 shows that architectural design affects viability through active decision quality rather than cost alone. CoT has the lowest total cost for both backbones, but it does not produce the best outcomes, suggesting that low-cost reasoning alone is insufficient for trading. The Appendix D.4 diagnosis further shows that this failure is not driven by cost accumulation and the largest negative contribution appears in the dynamic reallocation. By contrast, DeepFund achieves the strongest performance: it is the only architecture with positive net profit and agentic profit for DeepSeek-V3.2, and it also yields the highest values for GPT-5.2. Despite its higher cost, DeepFund's stronger observed performance mainly comes from its positive timing. It yields positive timing effects for both backbones, indicating more favorable dynamic reallocation under the evaluated setting. Thus, the value of an agentic architecture depends not on being cheaper or more complex, but on whether it can convert reasoning cost into better asset selection and timing decisions.

We further report complementary financial metrics (see Appendix D.6) to characterize the realized return and risk of the evaluated portfolios, together with an extended six-month evaluation using representative frontier backbones (see Appendix D.7).

6 Discussion

Sensitivity in Market Regime. Because trading performance is highly regime-dependent, we further examine viability and find that market regimes reshape profit sources more strongly than system costs, making attribution necessary for interpreting deployability across environments. Detailed results are provided in Appendix D.5.

Forward Diagnosis. We test whether **TradeLens** provides useful signals for later deployment decisions. Using the December diagnostic window, it flags hourly trading as cost- and friction-intensive, motivating a January 2026 daily-versus-hourly comparison. Although daily trading performs better in early January, both settings remain net negative, showing that cost control alone cannot ensure viability when strategy-level profit generation is weak. Detailed results are in Appendix E.1.

Practitioner Feedback. We also assess whether the diagnosis is interpretable and useful in private deployments. We collect feedback from 13 retail traders deploying **TradeLens**. Participants find the joint profit–cost diagnosis useful for distinguishing weak decision value, trading friction, and LLM usage. They also emphasize that actionable deployment support should go further by providing stronger strategy-level optimization guidance. Additional details are in Appendix E.2.

7 Conclusion and Future Work

We introduced **TradeLens**, a trace-grounded profit–cost viability diagnosis toolkit for evaluating agentic trading systems. It connects trading outcomes, runtime traces, and deployment costs under a shared accounting window, allowing users to assess whether a system is profitable as a whole and whether its LLM-mediated decisions generate enough incremental value to justify the induced costs. By distinguishing system from agentic viability, **TradeLens** diagnoses whether intelligence is converted into economic value. Future work will extend **TradeLens** to more asset classes, refine execution friction modeling, and optimize agent architectures toward sustainable economic returns.

Limitations

This work uses controlled experimental settings to demonstrate how **TradeLens** reveals profit–cost behaviors in agentic trading systems. Thus, the reported patterns should be viewed as diagnostic observations under specific settings, not general claims about the intrinsic trading ability of any backbone model. Their generalizability to other asset classes, broader investment universes, longer horizons, or different market regimes remains to be validated. We use the S&P 500 as a broad-market benchmark proxy in our experiments. This choice may introduce slight attribution bias in measuring systematic risk exposure, and users can customize the benchmark to better match their own trading universe and deployment needs. The practitioner evaluation involves a modest number of participants (13 retail traders) and collects qualitative feedback rather than controlled experimental evidence of downstream decision improvement. Additionally, cost inputs such as infrastructure fees and stochastic terms are user-configured rather than automatically measured; accuracy, therefore, depends on the fidelity of the values provided by the user.

Future Work

Future work can improve friction estimation, extend the toolkit to more asset classes and execution venues, and study how tool use, multi-agent coordination, and long-horizon reasoning affect the viability frontier. At the architecture level, controlled ablations could assess whether additional tool calls, specialized agents, or longer historical contexts provide incremental decision value sufficient to offset their computational and coordination costs. Such analyses could identify when additional reasoning is economically justified and when a simpler configuration offers comparable outcomes at lower cost.

We also plan to evaluate broader market regimes, incorporate finer-grained latency-to-fill logs and slippage models, and compare against stronger non-LLM baselines. For instance, evaluations spanning trending, sideways, and high-volatility markets could test whether observed profit sources remain stable as market conditions change. Notably, viability signals could be integrated with reinforcement learning (Jin et al., 2025; Li and Oliva, 2025) and human preference (Ong et al., 2025) to optimize agentic systems toward better economic outcomes. For example, training objectives could incorporate deployment costs and trading friction alongside returns and risk, while practitioner preferences could guide trade-offs among profitability, interpretability, and operational complexity.

Finally, these extensions would strengthen **TradeLens** as a framework for diagnosing and improving the economic viability of agentic trading systems.

Ethical Considerations

The present work has been conducted solely for academic research purposes and does not constitute investment advice or a recommendation to process any financial asset. Users deploying agentic trading systems in real markets should exercise judgment and comply with applicable regulations. We acknowledge that cost-aware evaluation, while promoting transparency, does not eliminate the risks inherent in algorithmic trading, including potential market impact, amplification of biases in LLM reasoning, and the possibility that cost-optimized configurations may inadvertently encourage excessive risk-taking.

All market data used in our experiments is publicly available third-party data accessed through licensed provider APIs, and we do not redistribute any raw data. The practitioner feedback study involved voluntary participants, and we collected only qualitative feedback without personally identifiable information, sensitive information, private trading strategies, or portfolio details. The diagnostic reports generated by **TradeLens** are intended as informational analysis tools and should not be interpreted as financial advice.

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A Diagnosis Report Format

In this part, we provide two formats to show what **TradeLens** outputs in practice; a complete output example is provided in our open source repository. First, we present an example of a financial report, which reports the exact profit and loss and other trading performance metrics of a trial run. Based on a detailed financial analysis report, **TradeLens** generates the diagnosis report with ranked suggestions from multiple dimensions, including capital budget, backbone model choice, trading frequency, and architectural intelligence such as modules, agents, and tools.

Financial Report Format

1. Trading Configuration
   - Trading Period:
   - Trading Model:
   - Assets:
2. Asset & Portfolio State
   - Cash condition:
   - Positions:
3. Performance
   - Profit Attribution and figure
   - Cost Breakdown and figure
   - Financial Statement
4. Execution Quality
   - Opportunity Cost and latency
   - Execution layer

Diagnosis Report Format

Executive Summary
Overview of the overall report and top suggestions

1. Trading Performance Analysis
   - Outcome Snapshot.
   - Profit Attribution Analysis.
   - Cost Structure Analysis.
   - Portfolio Timeline and Risk Signals.
   - Execution Quality and LLM Efficiency.
2. Trading System Diagnosis
   - System Diagnosis.
     Define core restrictive factors and auxiliary limiting conditions of strategy performance; assess the impact of fee expenditure on actual revenue.
   - Root Causes.
     Summarize fundamental issues behind unsatisfactory performance based on analytical data.
   - Top Recommended Actions.
     Provide recommendations targeting model parameters, capital allocation, and system framework; predict improvement effects and outline specific implementation approaches.
   - Quick Wins vs. Structural Changes.
     Short-term adjustable optimization measures vs. long-term systematic reconstruction and upgrade plans.

B Profit Attribution Details

As presented in Table 3, we attribute cumulative gross profit to three portfolio-level components. The decomposition is defined by two counterfactual baselines constructed on the same asset universe and evaluation window.

Component	Counterfactual	Economic interpretation
Market effect	Market return	Profit from exposure to aggregate market movement in the investable universe
Asset selection	Static initial allocation	Gain from the chosen initial weights relative to market exposure
Timing	Dynamic reallocation	Incremental gain from changing positions after the initial allocation

Table 3: Profit attribution categories for agentic trading systems.

Notation. Let $t \in \{1, \ldots, T\}$ index decision periods, N the number of tradable assets, $w_i^{(t)}$ the portfolio weight of asset i at period t with $\sum_i w_i^{(t)} \leq 1$, and $r_i^{(t)}$ its realized return. Let V_0 denote the initial portfolio value, $\alpha_0 = \sum_i w_i^{(0)}$ the initially invested fraction, and τ_0 the first period in which the agent establishes a non-cash allocation.

Systematic exposure baseline. Let r_t^{mkt} denote the realized market return in period t, reflecting aggregate market movement in the investable universe. In **TradeLens**, r_t^{mkt} is computed from the benchmark market prices supplied in the audit inputs. The counterfactual value under passive market exposure is

$$V_T^{\text{sys}} = V_0 \left[(1 - \alpha_0) + \alpha_0 \prod_{t=1}^{T} \left(1 + r_t^{\text{mkt}} \right) \right].$$

The attributed profit is $P^{\text{sys}} = V_T^{\text{sys}} - V_0$. Thus, an agent starting entirely in cash has zero Market Effect.

Static allocation baseline. The static baseline holds the agent's first active allocation, established at τ_0, unchanged for the remainder of the evaluation period:

$$V_T^{\text{base}} = V_0 \prod_{t=\tau_0}^{T} \left(1 + \sum_{i=1}^{N} w_i^{(\tau_0)} r_i^{(t)} \right).$$

The attributed profit is $P^{\text{asset}} = V_T^{\text{base}} - V_T^{\text{sys}}$. Hence, an agent may have zero Market Effect but non-zero Asset Selection if it starts in cash and establishes its first active allocation later.

Relation to classical attribution. Classical Brinson attribution compares portfolio and benchmark weights at the security level (Brinson et al., 1986, 1991). Our formulation operates at the portfolio-value level and uses two nested counterfactual paths rather than a single external benchmark. This design is tailored to agentic trading systems, where decisions are naturally described as market exposure, initial allocation, and subsequent reallocation, and where the evaluation goal is deployability diagnosing rather than performance ranking.

C Cost Taxonomy

Following Williamson's transaction cost economics (Williamson, 2005), markets are not frictionless: every exchange requires resources for information processing, decision-making, and execution, beyond price movement. Such transaction costs arise from bounded rationality and coordination actions under uncertainty, and they directly shape whether a strategy is viable in practice. This perspective is especially relevant for agentic trading systems, where "intelligence" is produced via paid inference and deployed through fee and impact sensitive execution pipelines. Motivated by this view, we construct an explicit cost model that covering end-to-end frictions across decision-making, trading execution, and infrastructure.

Let $t \in \{1, \ldots, T\}$ index decision periods within a time window T (e.g., minutes, hours, or days). We decompose the total cost in period t into LLM-side, trading-side, infra-side, and stochastic:

$$C_t = C_t^{\text{llm}} + C_t^{\text{trd}} + C_t^{\text{inf}} + C_t^{\text{sto}} \qquad (11)$$

Specifically, the components are computed as follows. The trading-side cost C_t^{trd} combines fixed charges and activity-dependent expenses,

$$C_t^{\text{trd}} = C_{\text{stat}}^{\text{trd}} + \sum_{i=1}^{M_{\text{trd}}} \gamma_{t,i}^{\text{trd}} \cdot \text{activity}_{t,i}^{\text{trd}}.$$

The LLM-side cost C_t^{llm} is determined by token usage and adjusted by the success rate,

$$c_t^{\text{llm}} = \frac{\sum_{s \in \mathcal{S}} p^s \cdot x_r^s}{\rho}, \qquad \mathcal{S} = \{\text{in, out, cache}\},$$

where ρ is successful completion rate due to retries.

The infra-side cost is modeled as a constant per period,

$$C_t^{\text{inf}} = \kappa,$$

and the stochastic cost captures unanticipated expenses,

$$C_t^{\text{sto}} \sim \text{Uniform}(0, n).$$

Given the cost taxonomy defined above, **TradeLens** leverages a *system-agnostic* cost accounting module that acts as a lightweight middleware between an agent's execution pipeline and the downstream accounting logic. Instead of assuming a specific prompting strategy or agent architecture, the taxonomy defines a small set of standardized events (with timestamps and identifiers) so that agentic workflows can all be logged and attributed consistently to a decision period and step. The dynamic portion of each category (*e.g.,* token-dependent inference fees, per-trade frictions such as commissions or slippage) is traced during execution through event logging. In contrast, the static portion (*e.g.,* data subscription or deployment baseline fees) is configured by end users as part of the evaluation setting, enabling **TradeLens** to reflect different billing plans and deployment assumptions without modifying the agent. Concretely, it considers the following aspects:

- **LLM-side accounting** C_t^{llm}: converts traced model usage (prompt/completion tokens, call counts, retries) into monetary inference cost under provider pricing (or user-specified prices).
- **Trading-side accounting** C_t^{trd}: aggregates execution-layer costs from orders and fills, including explicit fees (commissions, exchange/-clearing fees, taxes) and modeled frictions (spread/slippage) when measurable.
- **Infra-side accounting** C_t^{inf}: aggregates deployment costs such as elastic compute/bandwidth/I/O and monitoring/logging overhead; static baselines (*e.g.,* hosting plans) are incorporated from user configuration, while dynamic usage is traced during execution.
- **Stochastic term** C_t^{sto}: represents unanticipated or hard-to-measure expenses as an additive uncertainty term; it can be set to zero (best-case) or calibrated from real billing data (conservative-case) as discussed in previous work (Nevmyvaka et al., 2006; Gatheral, 2010).

As presented in Table 4, we categorize the cost items to four types according to (i) where the cost arises, and (ii) whether the cost is static (fixed for a period or billing cycle) or dynamic. As Stochastic term is sampled at runtime, minor cost differences may occur across independent runs.

D Experiment Detail

Across all experiments, we gathered data through an agentic trading system and recorded complete decision trajectories, including portfolio states and executed trades. We then apply **TradeLens** to perform profit–cost accounting and diagnostic analysis, allowing us to explain not only whether an agent is economically viable, but also why it succeeds or fails.

D.1 Detailed Experimental Setup

Cost assumptions. Specifically, we account for four **cost components**: (1) LLM-side: LLM token cost is computed from each model's usage and pricing from the official provider endpoints, and is adjusted by the task success rate (where ρ = 98% as described in LLM provider metrics) to reflect the effective cost per successful completion. (2) Trading-side: Commission fee is charged per execution, consistent with typical retail brokerage fees [1]. Thus, we calculate the relevant cost based on that documentation. (3) Infra-side: Infrastructure overhead is estimated at $0.20 per run, together with a fixed monthly data subscription fee (*e.g.,* $100/month [2]). Both are included in system-level deployment cost. (4) Stochastic: A stochastic cost term(*e.g.,* max. $0.5 per run) is added.

Profit attribution baselines. To complement the cost-side accounting, we construct two **profit-side baselines** for attribution analysis. The first is a systematic exposure baseline, which uses the return of the S&P 500 index over the same market window to capture broad market movement. In implementation, the market benchmark is applied only to the portion of capital initially invested in risky assets, while any uninvested capital remains in cash (Johnson, 2017; Brinson and Fachler, 1985). This preserves the agent's initial exposure decision and avoids implicitly assuming 100% market exposure when part or all of the initial capital is held as cash.

[1] https://www.interactivebrokers.com/en/pricing/commissions-stocks.php

[2] https://www.alphavantage.co/premium/

Category	Static (fixed)	Dynamic (variable)
LLM-side	Model subscription/licensing (if applicable)	Inference cost (prompt/completion tokens, context length, number of agent/-tool calls, retries)
Trading-side	Market data subscription; account/brokerage plan fees	Commissions/transaction fees; taxes (*e.g.,* stamp duty); exchange/clearing fees; bid–ask spread and slippage (market impact)
Infra-side	Hosting/deployment baseline; databases and storage	Elastic infrastructure usage (compute, bandwidth, I/O); monitoring/logging and reruns
Stochastic	Miscellaneous fixed charges not explicitly modeled	Residual/unanticipated costs (*e.g.,* unexpected platform fees, outages and reruns, sudden price changes)

Table 4: Cost categories for end-to-end accounting.

Consequently, the Market Effect may differ across backbone models if their initial invested fractions differ; a value of zero indicates no capital was initially exposed to risky assets. The second is a static allocation baseline, which follows a buy-and-hold strategy by keeping the initial portfolio allocation unchanged in the trading period.

Trading setup. The trading window is set to a two-month sideways market period from December 1, 2025 to January 30, 2026, starting with an initial cash of $100,000 and trading a fixed universe of the most liquid 10 liquid U.S. equities. We accessed market data via licensed provider APIs/subscriptions and do not redistribute the raw data. Asset universe: NVDA, AAPL, MSFT, AMZN, GOOGL, GOOG, META, AVGO, TSLA, and WMT.

D.2 Viability Across Capital Scale

Figure 4 shows that increasing capital scale does not lead to a uniform improvement in economic viability. Although larger capital may dilute fixed operating costs, the main variation in viability comes from profit-side components, especially asset selection and timing effects.

For GPT-5.2, system viability remains positive across all tested capital scales, although it does not increase monotonically with capital size. Net profit rises from 75.02 at 10k to 1041.22 at 50k, decreases to 246.97 at 100k, and then increases again to 1780.03 at 500k. However, agentic viability exhibits a different pattern. It is negative at 10k and 50k, becomes slightly positive at 100k, and then drops sharply to -5697.07 at 500k. Table 5 suggests that this deterioration at the largest tested scale is mainly driven by a strongly negative timing effect rather than by operating cost. Although GPT-5.2 remains system viable at 500k, its dynamic intervention fails to generate positive active profit after accounting for decision-induced costs.

Diagnosis sample: GPT-5.2 at 500k

Top Recommended Actions:
Reduce execution latency and slippage (System/agent/-tool architecture).
- Expected impact: rapidly recapture a large portion of the 3.98k opportunity cost and reduce timing erosion; could convert current fragile net profit into a robust positive.
- Implementation idea: move to a trimmed inference pipeline (pruned prompt/context, async batching, prioritized decisions), colocate order gateway or use faster broker API paths, and implement pre-validated order templates to eliminate per-trade orchestration delays. Measure by reducing the average latency per trade below 5s and the slippage by half.

DeepSeek-V3.2 shows a consistently weaker pattern. Both system and agentic viability remain negative across all capital scales. The losses are moderate at 10k–100k but expand substantially at 500k, where system viability decreases to -5679.66, and agentic viability decreases to -15873.79. Table 5 shows that this deterioration is not primarily caused by system cost, which remains small relative to profit variation. Instead, the largest losses come from unfavorable profit components, especially the strongly negative timing effect at 500k. These results suggest that capital scaling does more than dilute fixed costs. For GPT-5.2, market exposure and asset selection remain positive in aggregate across scales, while timing varies non-monotonically and becomes strongly negative at 500k.

Diagnosis sample: DeepSeek-V3.2 at 500k

- Token intensity is large (239k input tokens/day) — this implies expensive and heavy LLM usage and long prompt/context windows, increasing compute latency and opportunity cost.
- Distinction: selection skill is present (asset picks positive), but execution/system design (latency, LLM workflow) is producing missed fills and timing slippage. Market exposure (holding fast-rising assets) helped, but the system could not capture that upside because of when/how it traded.

Metric	DeepSeek-V3.2				GPT-5.2			
	10k	50k	100k	500k	10k	50k	100k	500k
Gross Profit [(a)]	-162.58	-491.32	-388.29	-5,425.95	303.15	1,278.89	505.76	2,022.49
Market Effect	16.38	829.56	1,351.45	7,715.41	169.57	858.88	1,535.80	8,130.57
Asset Selection	-114.91	-598.14	-1,500.44	2,686.92	292.92	526.37	-1,125.69	-445.27
Timing Effect [(1)]	-64.05	-722.73	-239.30	-15,828.28	-159.34	-106.36	95.65	-5,662.80
Total Cost [(b)]	266.72	300.54	304.94	253.71	228.13	237.66	258.79	242.46
Trading [(2)]	32.00	53.00	63.00	19.18	4.00	13.00	32.00	16.10
LLM [(3)]	15.03	29.55	23.84	15.58	4.52	7.37	9.03	6.72
Infra	208.20	208.20	208.20	208.20	208.20	208.20	208.20	208.20
Stochastic [(4)]	11.49	9.78	9.90	10.76	11.41	9.09	9.56	11.45
Net Profit [(a-b)]	-429.30	-791.85	-693.23	-5,679.66	75.02	1,041.22	246.97	1,780.03
Agentic Profit [(1-2-3-4)]	-122.57	-815.07	-336.04	-15,873.79	-179.27	-135.82	45.06	-5,697.07

Table 5: Viability across Capital scale, ranging from 10k USD to 500k USD, comparing DeepSeek-V3.2 and GPT-5.2. All values are rounded to two decimal places from raw data.

Decision Frequency	DeepSeek-V3.2		GPT-5.2	
	Hourly	Daily	Hourly	Daily
Gross Profit [(a)]	-906.86	197.23	-386.58	-51.59
Market Effect	239.70	190.59	241.95	216.59
Asset Selection	-473.75	233.02	-169.13	-73.62
Timing Effect [(1)]	-672.81	-226.38	-459.40	-194.56
Total Cost [(b)]	315.38	167.38	198.89	146.69
Trading [(2)]	92.00	45.00	19.00	32.00
LLM [(3)]	63.41	13.17	18.61	4.99
Infra	126.20	104.20	126.20	104.20
Stochastic [(4)]	33.78	5.01	35.08	5.50
Net Profit [(a-b)]	-1222.24	29.84	-585.46	-198.28
Agentic Profit [(1-2-3-4)]	-861.99	-289.56	-532.08	-237.05

Table 6: Viability across trading frequency.

For DeepSeek-V3.2, the large loss at 500k is dominated by negative timing despite positive asset selection. Thus, scaling can substantially change decision-attributed outcomes, and cost dilution alone is insufficient to ensure economic viability.

D.3 Viability Across Trading Frequency

Figure 5 shows that higher decision frequency does not improve viability. Compared with daily trading, hourly trading reduces gross profit by 1104.09 for DeepSeek-V3.2 and 334.99 for GPT-5.2, while the additional total cost accounts for only 11.8% and 13.5% of the corresponding net-profit deterioration. This indicates that the main failure mode is not fee burden, but degradation in trading decisions.

The attribution results, shown in Table 6, reveal that the degradation is concentrated in the active components.

For both backbones, hourly trading yields more negative timing effects than daily trading, suggesting that more frequent decision points add timing errors rather than stable short-term opportunities.

For DeepSeek-V3.2, the failure is more severe: hourly trading worsens both asset selection and timing effects, as frequent reallocation affects not only when the agent trades but also what it holds. Beyond the aggregate profit–cost accounting, we apply **TradeLens** to explain why higher decision frequency fails to improve viability. The diagnosis shows that the hourly setting does not merely increase operating cost, but also changes the quality of the active decisions. Thus, the hourly setting exposes an important issue: when the predictive signal is weak, increasing decision frequency amplifies noise, turnover, and mistimed reallocations instead of improving economic viability.

Diagnosis sample: DeepSeek-V3.2 under hourly trading

Trading cadence is likely too aggressive relative to predictive signal quality: 190 trades/month across multiple assets combined with negative timing indicates frequency is not justified by signal edge. Portfolio construction also contributed: initial holdings and sizing choices (see initial holdings versus final positions) show nontrivial exposures that did not capture market moves.

D.4 Viability Across System Architecture

Chain-of-Thought baseline. The CoT baseline uses the same market data and portfolio context as the agentic system, but removes tool interaction and multi-step external execution. For each trading date, the system first retrieves the current portfolio, daily open prices, previous open and close prices, and recent news for the watchlist.

It then constructs a single prompt that asks the LLM to analyze all tickers, provide a short portfolio-level synthesis, and output a final trading decision in a structured JSON format. Only the final JSON block is used for trade execution, while the preceding step-by-step analysis is logged as the model's reasoning trace.

Table 2 compares economic viability across different system architectures under the same initial capital of $100,000. The results show that the economic value of an architecture does not depend simply on whether it is cheaper or more complex, but on whether additional reasoning and coordination can be converted into better asset selection and timing decisions.

While CoT has the lowest total cost for both backbones, it fails to achieve the best economic outcome, proving that low operating costs alone do not guarantee viability. Weak reasoning structures still result in poor asset selection and ineffective reallocation despite lower expenses. This flaw is especially clear in DeepSeek-V3.2, where CoT caused a strongly negative timing effect and the worst overall agentic profit.

In contrast, DeepFund yields more stable gains across both backbones. It is the only architecture to achieve both positive net and agentic profit for DeepSeek-V3.2, while also capturing the highest net and agentic profit for GPT-5.2. Crucially, DeepFund does not have the lowest total cost. This proves its competitive advantage stems entirely from superior profit generation rather than cost reduction.

The attribution results further suggest that DeepFund's advantage is mainly associated with timing. It produces positive timing effects for both backbones, indicating that the architecture does not merely benefit from market exposure, but it actually also creates active value through dynamic reallocation. For retail investors interested in developing agentic trading systems, this suggests that the key challenge is not only to reduce inference cost or add more reasoning steps, but also to monitor complete trading trajectories and to verify whether reasoning cost is actually converted into better investment outcomes.

Diagnosis sample: DeepSeek-V3.2 under CoT

Dynamic cost accumulation is tiny relative to portfolio swings, so cumulative cost lines are almost flat while portfolio value moves dominate. The run exhibits path dependency driven by timing decisions: the timing effect (-3,214.75) shows losses are concentrated in when trades were executed, not in cost build-up.
- Fragility of profitability:
- Given small cost base and positive selection when capital is invested, profitability is fragile but fixable — i.e., the strategy can produce positive selection returns if timing is corrected or exposure windows are altered.
- However, until timing decisions improve, increasing capital or leaving the static subscription in place risks repeated net losses.

D.5 Sensitivity in Market Regime

Market regime is typically discussed for its impact on returns (Ang and Timmermann, 2012; Dong et al., 2025). For agentic trading systems, regime shifts may also affect viability by changing both profit opportunities and deployment frictions. For example, liquidity and volatility can alter execution costs such as spread and slippage, while different market conditions may induce different trading intensity and runtime workload. Thus, before attributing net-profit changes to model or system design alone, we test whether profit–cost viability is sensitive to market regimes.

In order to isolate temporal and regime effects, we conduct several additional experiments under bearish (2025-02-10 to 2025-04-07), sideways (2025-12-01 to 2026-01-30), and bullish (2025-06-02 to 2025-08-08) conditions. The results are shown in Table 7 , which shows that profit varies much more sharply than cost across regimes. For both DeepSeek-V3.2 and GPT-5.2, total costs remain within a relatively narrow range, whereas gross profit, net profit, and agentic profit change substantially with market conditions.

Metric	DeepSeek-V3.2			GPT-5.2		
	Bearish	**Sideways**	**Bullish**	**Bearish**	**Sideway**	**Bullish**
Gross Profit [(a)]	-21770.56	-388.30	11249.93	-27966.93	505.76	14887.99
Market Effect	-2488.54	1351.45	755.35	-16570.14	1535.80	4294.88
Asset Selection	-2328.65	-1500.44	272.15	-10920.83	-1125.69	6735.38
Timing Effect [(1)]	-16953.37	-239.30	10222.43	-475.96	95.65	3857.73
Total Cost [(b)]	281.89	304.94	274.08	236.11	258.79	286.83
Trading [(2)]	39.00	63.00	29.00	12.00	32.00	58.00
LLM [(3)]	26.24	23.84	23.85	7.53	9.03	9.74
Infra	207.80	208.20	209.40	207.80	208.20	209.40
Stochastic [(4)]	8.84	9.90	11.83	8.77	9.56	9.69
Net Profit [(a-b)]	-22052.45	-693.23	10975.85	-28203.04	246.97	14601.16
Agentic Profit [(1-2-3-4)]	-17027.46	-336.04	10157.75	-504.27	45.06	3780.30

Table 7: Viability across market conditions. Cumulative profit and cost breakdown with an initial cash of $100,000.

In the bearish regime, both models are system-non-viable, but the failure mechanisms differ. DeepSeek-V3.2 suffers from a large negative timing effect, indicating that dynamic reallocation amplifies downside exposure. GPT-5.2, by contrast, is mainly hurt by negative market exposure and asset selection, while its timing loss is relatively small. In the sideways regime, GPT-5.2 achieves modest positive net profit and agentic profit, whereas DeepSeek-V3.2 remains negative, suggesting that the same market condition can still expose backbone-specific differences in timing and execution quality. In the bullish regime, both models become strongly profitable, but through different profit sources: DeepSeek-V3.2 is dominated by timing gains, while GPT-5.2 benefits from both asset selection and timing. Overall, the market regime mainly determines whether LLM-mediated decisions can yield profit. This result reinforces the need for profit–cost diagnosis: similar cost levels can correspond to very different viability outcomes because market exposure, asset selection, and timing respond differently across regimes.

D.6 Complementary Financial Performance Metrics

We additionally report conventional financial metrics as complementary diagnostics for the evaluated trading configurations. Agentic viability is not intended to replace these standard measures: conventional metrics characterize the realized performance and risk profile of the overall portfolio, whereas our analysis focuses on whether the dynamic LLM-mediated layer generates sufficient incremental value to justify its costs.

Total Return. Total return measures the cumulative change in portfolio value over the evaluation period:

$$R = \frac{V_T - V_0}{V_0},$$

where V_0 and V_T denote the initial and final portfolio values, respectively.

Sharpe Ratio. The annualized Sharpe ratio is computed from daily portfolio returns:

$$S = \frac{\bar{r} - r_f}{\sigma_r}\sqrt{252},$$

where $\bar{r}$ is the average daily return, r_f is the daily risk-free rate, and σ_r is the standard deviation of daily returns.

Maximum Drawdown. Maximum drawdown measures the largest decline from a historical portfolio-value peak:

$$\mathrm{MDD} = \max_t \left(\frac{V_t^{\max} - V_t}{V_t^{\max}} \right),$$

where

$$V_t^{\max} = \max_{\tau \le t} V_\tau.$$

Turnover. Turnover measures the amount of trading relative to portfolio value:

$$\mathrm{TO} = \frac{1}{T} \sum_{t=1}^{T} \frac{A_t}{V_t},$$

where A_t denotes the total traded amount at trading step t, V_t is the portfolio value, and T is the number of trading steps.

Model	Total Return(%)	Sharpe S	Max DD(%)	Turn-over
Qwen3-Max	-0.05	-5.81	0.22	0.08
Mistral-Large-3	0.78	0.13	4.71	5.06
MiniMax-M2.1	-1.47	-0.70	6.73	2.21
Llama-4-scout	-1.02	-0.71	4.9	0.92
Kimi-K2	-1.98	-1.06	5.84	2.13
GPT-5.2	0.51	0.02	4.79	1.16
GLM-4.7	-2.55	-1.30	6.44	5.92
Gemini 3 Flash	-0.98	-0.55	7.06	5.67
DeepSeek-V3.2	-0.39	-0.42	4.96	2.37
Claude Sonnet 4.5	0.52	0.03	5.69	1.33

Table 8: Standard financial metrics for the model comparison experiment with $100K initial capital and daily trading frequency (time window: Dec. 2025–Jan. 2026).

Model	Capital	Total Return(%)	Sharpe	Max DD(%)	Turn-over
DeepSeek-V3.2	$10k	-1.63	-0.98	5.82	2.39
	$50k	-0.98	-0.67	4.89	1.97
	$100k	-0.39	-0.42	4.9	2.37
	$500k	-1.09	-0.97	5.49	1.38
GPT-5.2	$10k	3.03	1.04	5.41	0.98
	$50k	2.56	0.82	5.06	1.39
	$100k	0.51	0.02	4.79	1.16
	$500k	0.4	-0.02	5.39	1.13

Table 9: Standard financial metrics for the capital scale experiment with daily trading frequency (time window: Dec. 2025–Jan. 2026).

Tables 8, 9, 10, 11, and 12 report total return, Sharpe ratio, maximum drawdown, and turnover for the experimental settings in Section 5.2 and Appendix D.5.

D.7 Extended Evaluation

To further examine whether the findings are specific to the original two-month evaluation window, we conduct an extended six-month experiment using three representative frontier backbone models: GPT-5.4(OpenAI, 2026), DeepSeek-V4-Flash(DeepSeek-AI et al., 2026), and Gemini 3.5 Flash(Kavukcuoglu et al., 2026). Each agent starts with $100K initial capital and makes daily trading decisions from January 1st to July 1st 2026(SPY returns 9.36%, 124 trading days).

Table 13 reports both conventional financial metrics and TradeLens attribution results. GPT-5.4 is the only model that achieves a positive net return, while all agents underperform the SPY benchmark. Across models, the positive market component is offset by negative selection value, while timing differs substantially across agents: GPT-5.4 partially recovers selection losses through positive timing, whereas DeepSeek-V4-Flash and Gemini 3.5 Flash incur additional losses from dynamic trading.

Model	Freq	Total Return(%)	Sharpe	Max DD(%)	Turn-over
DeepSeek-V3.2	daily	0.2	-0.06	3.77	1.93
	hourly	-0.91	-0.62	4.12	3.36
GPT-5.2	daily	-0.05	-0.28	3.68	1.16
	hourly	-0.39	-0.05	5.86	1.34

Table 10: Standard financial metrics for the trading frequency experiment (time window: Dec. 2025).

Model	Variant	Total Return(%)	Sharpe	Max DD(%)	Turn-over
DeepSeek-V3.2	CoT	-1.24	-0.70	5.18	2.83
	DeepFund	0.61	-0.02	2.65	6.87
GPT-5.2	CoT	0.34	-0.04	4.55	1.82
	DeepFund	0.88	0.06	3.46	5.58

Table 11: Standard financial metrics for the system complexity experiment (time window: Dec. 2025–Jan. 2026).

These results suggest that extending the evaluation horizon does not eliminate the central viability issue: realized portfolio performance depends not only on market exposure, but also on whether dynamic agentic decisions generate sufficient incremental value relative to their associated costs.

E Deployment User Study

This part jointly evaluates deployability from both technical and user-facing perspectives: whether **TradeLens** can provide forward-looking deployment diagnostics, and whether practitioners find such diagnostics useful for analyzing and revising agentic trading systems.

E.1 Forward Diagnostic Analysis

Diagnostic finding in December 2025. Under the hourly configuration, the testing system starts with $100,000 and ends in December with a gross loss of approximately $900, while incurring about $300 in total system cost.

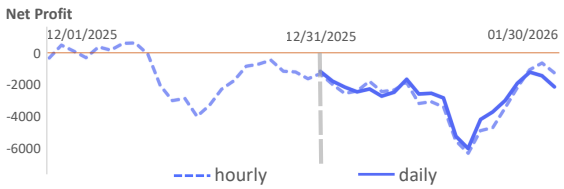


Figure 6: Forward diagnostic analysis of trading frequency. Cumulative net profit of DeepSeek-V3.2 under hourly (dashed line) and daily (solid line) trading frequencies. The vertical gray line marks the transition date.

Model	Window	Condition	Total Return (%)	Sharpe	Max DD(%)	Turnover
DeepSeek-V3.2	2025-02 ∼2025-04	bearish	-21.77	-5.72	22.21	1.49
	2025-06 ∼2025-08	bullish	11.25	3.76	3.51	1.4
GPT-5.2	2025-02 ∼2025-04	bearish	-27.97	-5.96	28.7	1.20
	2025-06 ∼2025-08	bullish	14.89	4.56	3.67	1.98

Table 12: Standard financial metrics for the market regime experiment.

Model	Total Return (%)	Sharpe	Max DD (%)	Turnover	Gross ($)	Market	Selection	Timing	Total Cost ($)	Net Profit ($)	Agentic Profit ($)
GPT-5.4	5.08	0.37	14.75	18.64	5,080	8,754	−9, 328	5,655	873	4,207	5,407
DeepSeek-V4-Flash	−6.23	−0.61	18.28	13.77	−6, 228	9,192	−9, 558	−5, 862	1,068	−7, 296	−6, 305
Gemini 3.5 Flash	−11.31	−0.62	28.35	163.93	−11, 310	9,290	−18, 551	−2, 048	1,040	−12, 350	−2, 464

Table 13: Extended evaluation of three representative frontier backbone models. All agents start with $100K initial capital and make daily trading decisions from January to July 2026. SPY returns 9.36% over the same period.

Fixed data subscription fees account for roughly $100, and the main dynamic costs come from trading-side and LLM-side usage. The diagnostic report further identifies excessive token usage per decision and concentrated capital utilization. Based on these accounting signals, **TradeLens** flags trading frequency as a major cost driver and recommends reducing decision frequency as a high-priority adjustment.

Forward evaluation in January 2026. We evaluate the frequency-reduction suggestion in a later market window by comparing the daily configuration with hourly baseline. In January, the daily configuration reduces total system cost from approximately $610 under the hourly baseline to about $124, and achieves higher net profit in early January (Figure 6). However, net profit remains negative under both configurations. A reversal occurs on January 28, when the daily configuration abstains from trading under unfavorable market conditions, whereas the hourly configuration continues active trading and partially mitigates losses. By the end of January, the hourly baseline records a smaller net loss than the reduced-frequency configuration.

These results show that the December diagnosis correctly identifies trading frequency as an important cost driver, but cost reduction alone does not guarantee later-window viability. When gross returns are constrained by market conditions and strategy-level edge, lowering decision frequency can reduce cost while still failing to improve final net profit. Thus, the forward analysis supports the role of **TradeLens** as a diagnostic tool: it explains why a system fails to become viable, rather than assuming that a cost-saving suggestion will always improve performance.

ID	Background	Trading Exp.	AI Exp.
P1	Undergraduate Student	1–3 yrs	Low
P2	Graduate Student	1–3 yrs	Medium
P3	Individual Investor	3–5 yrs	Medium
P4	Individual Investor	3–5 yrs	Low
P5	Finance Practitioner	5+ yrs	Medium
P6	Graduate Student	1–3 yrs	Low
P7	Data Analyst	3–5 yrs	Medium
P8	AI System Developer	5+ yrs	High
P9	Individual Investor	1–3 yrs	Medium
P10	Software Engineer	3–5 yrs	Low
P11	Finance Practitioner	5+ yrs	Medium
P12	Undergraduate Student	1–3 yrs	Low
P13	Retail Trading Enthusiast	3–5 yrs	Medium

Table 14: Participant demographics for the deployment-oriented usability assessment.

E.2 Practitioner Feedback

Participants. In order to complement the forward diagnostic analysis, we further collected practitioner's feedback on whether **TradeLens** helps users interpret the deployability of private agentic trading systems. We recruited a total of thirteen retail traders who used **TradeLens** in their own agentic trading workflows while keeping their strategies, asset universes, trading logs, and portfolio details private. The participants were recruited through social networks, and a startup company specializing in financial AI, who agreed to support this effort. Several employees of the company voluntarily participated in the experiment and were allowed to withdraw at any time. To preserve their anonymity, we do not disclose the company name. In this collaboration, we collected only participants' relevant professional experience and feedback, and did not collect any personally identifiable information. Participant demographics are summarized in Table 14.

Procedure. In order to ensure valid participation, we carefully explained the purpose of the data collection and how the collected data would be used afterwards, and we obtained informed consent from all participants before any data collection has started. The participants then received a brief usage guide explaining the required inputs, profit–cost metrics, and report structure. They then used **TradeLens** in order to audit profit sources, cost drivers, net profit, system viability, and agentic viability for their own runs. After using the toolkit, they completed a 5-point Likert questionnaire and provided open-ended comments. The questionnaire covered the following three dimensions: (i) *profit–cost diagnosis*, including whether the report helped users understand viability, distinguish market exposure, asset selection, and timing, identify dominant cost drivers, and separate weak decision value from high cost, (ii) *diagnostic actionability*, including whether the report supported revision decisions such as model choice, trading frequency, architecture, or execution adjustment, and (iii) *usability and integration*, including whether the outputs were understandable and whether the integration effort was acceptable.

Main feedback. Overall, the participants generally found **TradeLens** to be quite useful for diagnosing why a private trading agent was or was not deployable, instead of merely reporting its final profit. They emphasized that profit and cost should be interpreted jointly, since poor net performance may result from different underscoring mechanisms, including weak asset selection, mistimed reallocation, excessive trading friction, high LLM usage, or fixed data and infrastructure fees. The participants further found the distinction between system viability and agentic viability helpful, because it separates systems that merely benefit from market exposure from those whose dynamic decisions add incremental value.

From a developer's perspective, participants highlighted three practical benefits. First, the reports made debugging more traceable by linking failure modes to trading records, runtime traces, and cost components. Second, the configuration-level recommendations were actionable, such as reducing trading frequency, limiting redundant model calls, or improving execution design. Third, the reports helped prioritize whether a failure should be addressed through operational tuning or deeper architectural revision.

Participants also noted several limitations. They requested more concrete strategy-level guidance after failure modes are identified, especially for asset selection, timing adjustment, and position control. Some participants suggested making benchmark choice, execution-cost assumptions, and fixed infrastructure fees more explicit, so that the diagnosis can be better adapted to different deployment settings. These results support **TradeLens** as a practical tool for assessing whether LLM-mediated decisions convert induced costs into deployable trading value under realistic private deployment constraints.

Ethics statement. This feedback study was reviewed and approved by the institutional ethics committee of our university. All participants were informed of the purpose of the study before providing feedback. Participation was voluntary, and participants could withdraw at any time. The collected feedback was used only for research purposes, and all demographic information was anonymized and reported in aggregate form.